

2024, Mar. 2024 Release LOC:Acute Pediatric**Cystic Fibrosis****Utilization Benchmarks****Length of Stay:** 9.6 d**Length of Stay Type:** InterQual**Condition/Procedure:** CYSTIC FIBROSIS**% Paid as Observation:** 49%

Percent Paid as Observation indicates the final payment status as a percentage for patients hospitalized with the condition. It may not correspond to the status assigned at admission. Note that patients initially placed in Observation status who are converted to inpatient status (common) are included in the inpatient category, and that patients initially admitted to inpatient status who are converted to observation status (uncommon) are included in the Observation category.

Overview**Select Day**

Episode Day 1

Episode Day 2-5

Episode Day 6

Episode Day 7-9

Episode Day 10

Discharge Screens ⁽⁷³⁾**Notes**

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Overview**Introduction:**

Cystic fibrosis is a multisystem disorder caused by a genetic mutation that produces increased sodium levels in combination with ineffective chloride absorption creating thickened mucus. Complications of mucus production include the obstruction of the pancreatic ducts, sinus tracts, and airways. Antibiotic therapy usually consists of a 10-14 day duration and is based on response to treatment. Severe exacerbations or prolonged symptom recovery may require additional treatment up to 21 days. Exacerbations that are treated appropriately prevent further damage to the lungs and have better survival rates. The rate of decline in lung function is associated with the frequency and severity of pulmonary exacerbations, defined as the acute onset of respiratory impairment. The etiology of a pulmonary exacerbation is multifactorial and may include: decreased immunity, changes in airway microbiology (e.g., invasion by new organisms or strains of existing organisms, increased concentration of colonizing flora, or growth of existing strains), inflammatory response, and poor compliance with prescribed maintenance therapy. While outpatient management of pulmonary exacerbations may be an option for select patients, hospitalization is recommended when outpatient resources cannot match the close observation and intensity of therapy required in the inpatient setting.

Evaluation/Treatment:

Currently, no accepted criteria exist to confirm a pulmonary exacerbation. Diagnosis is based on a constellation of clinical signs and symptoms including: changes in cough, sputum production, work of breathing, energy level, appetite, weight loss, and pulmonary function tests. Treatment consists of intravenous anti-infective therapy and intensification of pulmonary clearance measures. Anti-infective therapy should be based on prior culture results, with modification determined by new culture results and clinical response. Chronic infection is the persistent isolation of an organism on repeat cultures over 6 months to a year leading to a high rate of acquired antimicrobial resistance. In addition to anti-infective treatment, guidelines recommend continuation of baseline medication regimens and intensification of airway clearance therapy in the overall management of a pulmonary exacerbation. The treatment regimen should also address the common comorbidities associated with cystic fibrosis:

- Malnutrition: Increased energy expenditure during a pulmonary exacerbation requires optimization of nutrition via pancreatic enzyme replacement therapy and a high calorie, high protein diet.
- Cystic Fibrosis-Related Diabetes: Insulin requirements may increase during a pulmonary exacerbation.
- Constipation: Adherence to a strict bowel regimen in patients with impaired gastrointestinal motility is necessary to prevent obstruction.
- Renal disease: Chronic use of nephrotoxic antibiotics (e.g., aminoglycosides) contributes to long-term renal damage. Close monitoring of renal function is recommended in patients receiving intravenous aminoglycosides.

Reference: (Talwalkar and Murray, Clin Chest Med 2016, 37: 69-81; Gillespie et al., Int J Hepatol, 2017: 5128760; Bos et al., J Cyst Fibros 2017, 16: 13-23; Balaguer and Gonzalez de Dios, Cochrane database of systematic reviews 2012, 3: CD001917)

In an effort to address inequities in medicine and improve appropriate access to care, multiple medical societies have been reevaluating the use of adjustments made for a patient's race and updating their calculators and reference equations when reporting diagnostic results.

In 2023, the American Thoracic Society issued updated guidance on reporting of pulmonary function tests (PFTs), advocating for a race-neutral average reference equation to determine the baseline (i.e., predicted normal) for a given patient (Bhakta et al., Am J Respir Crit Care Med 2023, 207: 978-995). The reference equation previously included lower predicted normal values for patients of certain races and ethnicities (in the US, for Black and Asian patients); a practice which has not been found to have biological validity and may lead to misclassification of diagnoses and disease severity in racial or ethnic minorities (Vyas et al., N Engl J Med 2020, 383: 874-882).

Since not all spirometer manufacturers and PFT laboratories have switched over to the new race-neutral average reference equation, clinical judgement and consideration of patient-specific factors should be used when interpreting PFT results and deciding whether patients with borderline reported PFT values in the presence of concerning symptoms or signs should have further evaluation or treatment.

(Excludes PO medications unless noted)

Select Day, One:● **Episode Day 1, One:**

(Symptom or finding within 24h)

● **ACUTE, One:** ⁽¹⁾● Pulmonary exacerbation, **Both:** ^(2, 3)● Finding, ≥ **Two:**

- Cough and increased intensity or frequency
- Decreased exercise tolerance

● Failed outpatient treatment and, **All:** ⁽⁴⁾

- Anti-infective (new)
- Nebulizer treatment with increased dose or frequency
- No evidence of clinical improvement ⁽⁴⁾
- Hemoptysis, increased or new onset ⁽⁵⁾

● Hypoxia and, ≥ **One:** ⁽⁶⁾

- New supplemental oxygen requirement
- Increased supplemental oxygen requirement
- O₂ sat decreased by > 4%(0.04) from baseline ⁽⁷⁾

● Increased work of breathing, **Both:**● Respiratory rate, **One:**

- Respiratory rate > 40/min, sustained (age ≤ 12 yrs) ⁽⁸⁾
- Respiratory rate > 30/min, sustained (age > 12 to < 18 yrs) ⁽⁸⁾

● Symptom, ≥ **One:**

- Dyspnea, worsening
- Use of accessory muscles
- Retractions
- New pulmonary infiltrate by chest x-ray (CXR)

● Pulmonary function, **Both:**

- FEV₁ decreased by ≥ 10%(0.10) from baseline ^(9, 10)
- FEV₁ performed within last 3 months ⁽⁹⁾
- Sputum production increased
- Temperature ≥ 100.4°F(38.0°C) ⁽¹¹⁾
- Weight loss ⁽¹²⁾
- Wheezing, new onset

● Intervention, **Both:**● Airway clearance therapy, **One:** ⁽¹³⁾

- At least 3x/24h
- Contraindicated due to hemoptysis ⁽¹⁴⁾
- Anti-infective

● **INTERMEDIATE, Both:** ⁽¹⁵⁾● Respiratory insufficiency, ≥ **One:**

- Arterial Po₂ 40-54 mmHg(5.3-7.2 kPa)
- Arterial Pco₂ 50-59 mmHg(6.7-7.9 kPa) and pH < 7.35 without mental status change ⁽¹⁶⁾
- Venous Pco₂ 50-59 mmHg(6.7-7.9 kPa) and pH < 7.35 without mental status change ⁽¹⁶⁾

● O₂ sat, **One:**

- < 90%(0.90) and < baseline ⁽⁷⁾
- Room air assessment not clinically appropriate ⁽¹⁷⁾

● Intervention, **Both:** ^(3, 12)

- Anti-infective

● Oxygen, **One:**● Respiratory intervention, **Both:** ^(13, 18)● Intervention, ≥ **One:**

- HFNC and, ≥ **One:** ⁽¹⁹⁾

- Flow rate, **One:** ⁽²⁰⁾
 - ≤ 20 L/min (> 10 kg)
 - < 2 L/kg/min (≤ 10 kg)
 - $\text{FiO}_2 > 21\%(0.21)$ to $39\%(0.39)$ ⁽²¹⁾
 - Oxygen $35\text{--}39\%(0.35\text{--}0.39)$ ⁽²¹⁾
 - Respiratory interventions every 3-4h ^(13, 18)
 - NIPPV ⁽²²⁾
- **CRITICAL, $\geq$ One:** ⁽²³⁾
 - Hemodynamic instability, **Both:**
 - Finding, $\geq$ **Two:**
 - Heart rate, **One:** ⁽²⁴⁾
 - > 180 /min, sustained (age ≤ 12 mos) ⁽⁸⁾
 - > 160 /min, sustained (age > 12 mos to 4 yrs) ⁽⁸⁾
 - > 150 /min, sustained (age > 4 yrs to 8 yrs) ⁽⁸⁾
 - > 140 /min, sustained (age > 8 yrs to 15 yrs) ⁽⁸⁾
 - > 130 /min, sustained (age > 15 yrs to < 18 yrs) ⁽⁸⁾
 - Systolic BP, **One:**
 - < 70 mmHg (age ≤ 12 mos)
 - < 80 mmHg (age > 1 to < 5 yrs)
 - < 90 mmHg (age 5 to < 18 yrs)
 - Labile ⁽²⁵⁾
 - MAP, **One:** ⁽²⁶⁾
 - < 55 mmHg (≤ 12 mos)
 - < 65 mmHg (> 1 to < 18 yrs)
 - Intervention, $\geq$ **One:**
 - Antifibrinolytic ⁽²⁷⁾
 - Blood product transfusion ⁽²⁸⁾
 - IV fluid resuscitation ≤ 2 d ⁽²⁹⁾
 - Hemodynamic monitoring, invasive ⁽³⁰⁾
 - Hemoptysis and bronchial artery embolization ⁽³¹⁾
 - HFNC and, $\geq$ **One:** ⁽¹⁹⁾
 - Flow rate, **One:** ⁽³²⁾
 - > 20 L/min (> 10 kg)
 - ≥ 2 L/kg/min (≤ 10 kg)
 - $\text{FiO}_2 \geq 40\%(0.40)$ ⁽²¹⁾
 - Mechanical ventilation
 - NIPPV and, $\geq$ **One:** ⁽²²⁾
 - Arterial $\text{Po}_2 \leq 39$ mmHg(5.2 kPa)
 - Arterial or venous $\text{Pco}_2 \geq 60$ mmHg(8.0 kPa)
 - Arterial or venous pH ≤ 7.24
 - Oxygen $\geq 40\%(0.40)$ ⁽²¹⁾
 - Vasopressor, vasodilator, or inotrope continuous (excludes low or fixed dose for end stage heart failure)
- **Episode Day 2-5, One:**
 - **ACUTE, Both:**
 - Airway clearance therapy, **One:** ⁽¹³⁾
 - At least 3x/24h
 - Contraindicated due to hemoptysis ⁽¹⁴⁾
 - Anti-infective
 - **INTERMEDIATE, $\geq$ One:**
 - Anti-infective desensitization ≤ 24 h ⁽³³⁾
 - HFNC and, $\geq$ **One:** ^(19, 34)
 - Flow rate, **One:** ⁽²⁰⁾
 - ≤ 20 L/min (> 10 kg)

- < 2 L/kg/min (≤ 10 kg)
- $\text{FiO}_2 > 21\%(0.21)$ to $39\%(0.39)$ ⁽²¹⁾
- NIPPV ⁽²²⁾
- Oxygen $35\text{-}39\%(0.35\text{-}0.39)$ ⁽²¹⁾

● **CRITICAL, $\geq$ One:**

- Acute kidney injury (AKI) and, **Both:** ⁽³⁵⁾

- Finding, $\geq$ **One:**

- Potassium > 6.0 mEq/L(6.0 mmol/L) and, $\geq$ **One:**

- Loss of P wave
 - Multifocal PVCs
 - Neuromuscular deficit ⁽³⁶⁾
 - Widening QRS

- Sodium, **One:**

- < 120 mEq/L(120 mmol/L) and volume overload
 - > 158 mEq/L(158 mmol/L)

- Intervention, $\geq$ **One:**

- Calcium chloride
 - Calcium gluconate
 - Glucose $50\%(0.50)$ with insulin ⁽³⁷⁾
 - Diuretic, continuous
 - IV fluid resuscitation $\leq 2\text{d}$ ⁽²⁹⁾

- Dialysis, **One:**

- Hemodialysis $\leq 7\text{d}$ since initiation
 - Peritoneal dialysis $\leq 7\text{d}$ since initiation

- CRRT ⁽³⁸⁾

- DIC and blood product transfusion ⁽³⁹⁾

- ECMO or ECLS ^(40, 41)

- Hemodynamic instability, **Both:**

- Finding, $\geq$ **Two:**

- Heart rate, **One:** ⁽²⁴⁾

- > 180/min, sustained (age ≤ 12 mos) ⁽⁸⁾
 - > 160/min, sustained (age > 12 mos to 4 yrs) ⁽⁸⁾
 - > 150/min, sustained (age > 4 yrs to 8 yrs) ⁽⁸⁾
 - > 140/min, sustained (age > 8 yrs to 15 yrs) ⁽⁸⁾
 - > 130/min, sustained (age > 15 yrs to < 18 yrs) ⁽⁸⁾

- Systolic BP, **One:**

- < 70 mmHg (age ≤ 12 mos)
 - < 80 mmHg (age > 1 to < 5 yrs)
 - < 90 mmHg (age 5 to < 18 yrs)
 - Labile ⁽²⁵⁾

- MAP, **One:** ⁽²⁶⁾

- < 55 mmHg (≤ 12 mos)
 - < 65 mmHg (> 1 to < 18 yrs)

- Intervention, $\geq$ **One:**

- Antifibrinolytic ⁽²⁷⁾
 - Blood product transfusion ⁽²⁸⁾
 - IV fluid resuscitation $\leq 2\text{d}$ ⁽²⁹⁾

- Hemodynamic monitoring, invasive ⁽³⁰⁾

- Hemoptysis $> 240\text{ml}/24\text{h}$ and, $\geq$ **One:** ⁽⁴²⁾

- Lung resection required (**use General Surgical subset**) ⁽⁴³⁾
 - Blood product transfusion ⁽²⁸⁾
 - Bronchial artery embolization performed within last 24h ⁽³¹⁾
 - Post bronchial artery embolization monitoring $\leq 24\text{h}$

- HFNC and, **≥ One:** ^(19, 34)
 - Flow rate, **One:** ⁽³²⁾
 - > 20 L/min (> 10 kg)
 - ≥ 2 L/kg/min (≤ 10 kg)
 - FiO₂ ≥ 40%(0.40) ⁽²¹⁾
 - Mechanical ventilation
- NIPPV and, **≥ One:** ⁽²²⁾
 - FiO₂ ≥ 40%(0.40)
 - Respiratory intervention every 1-2h ⁽¹⁸⁾
- Oxygen ≥ 40%(0.40) ⁽²¹⁾
- Vasopressor, vasodilator, or inotrope continuous (excludes low or fixed dose for end stage heart failure)
- **Episode Day 6, One:**
 - **ACUTE, One:**
 - **Early responder**, discharge expected today if clinically stable last 24h, **All:** ⁽⁴⁴⁾
 - Anti-infective regimen, **One:**
 - Complete
 - Established and tolerated, **One:**
 - Transition to PO anti-infective
 - Vascular access established for IV anti-infective
 - O₂ sat ≥ 94%(0.94) or within acceptable limits ⁽⁴⁵⁾
 - Pulmonary function return to baseline or improving ^(7, 46)
 - Tolerating PO or nutritional route established ^(47, 48)
 - Treatment needs manageable in post acute setting ⁽⁴⁹⁾
 - No complication or active comorbidity relevant to this episode of care ⁽⁵⁰⁾
 - **Partial responder**, not clinically stable for discharge and requires continued stay, **≥ One:**
 - Inadequate clinical improvement and, **Both:** ⁽⁴⁶⁾
 - Finding, **≥ One:**
 - FEV₁ < 80%(0.80) of baseline ^(9, 10)
 - Hemoptysis, persistent
 - O₂ sat < 94%(0.94) and < baseline requiring supplemental oxygen ^(7, 21)
 - Pulmonary infiltrates worsening by imaging
 - Sputum production increasing
 - Intervention, **Both:**
 - Airway clearance therapy, **One:** ⁽¹³⁾
 - At least 3x/24h
 - Contraindicated due to hemoptysis ⁽¹⁴⁾
 - Anti-infective
 - IV anti-infective regimen incomplete and, **Both:**
 - Barrier to discharge, **≥ One:** ⁽⁴⁹⁾
 - Patient or caregiver unable to adhere to treatment regimen
 - Treatment intensity exceeds resources available in post acute setting
 - Intervention, **Both:**
 - Airway clearance therapy, **One:** ⁽¹³⁾
 - At least 3x/24h
 - Contraindicated due to hemoptysis ⁽¹⁴⁾
 - Anti-infective
 - AKI, **Both:** ⁽³⁵⁾
 - Finding, **≥ One:**
 - Oliguria ^(51, 52)
 - Creatinine, **One:** ⁽⁵³⁾
 - ≥ 1.5x baseline and chronic kidney disease ⁽⁷⁾
 - ≥ 1.5x ULN
 - GFR > 25%(0.25) decrease from baseline ^(7, 54)

- Dialysis initiated ⁽⁵⁵⁾
- **Intervention, ≥ One:**
 - Volume expander ≤ 2d ⁽⁵⁶⁾
 - Diuretic ≥ 2x/24h, ≤ 3d
 - Dialysis (initial) ≤ 7d since initiation ^(55, 57)
 - Dialysis discontinued requiring lab monitoring ≤ 2d since discontinuation ⁽⁵⁸⁾
 - Immunosuppressant therapy ≤ 3d ^(59, 60)
- Adverse drug reaction requiring change in medication ≤ 24h ⁽⁶¹⁾
- Anti-infective level non-therapeutic and adjustment in dose or frequency ≤ 2d
- **Blood glucose ≤ 24h, Both:** ⁽⁶²⁾
 - **Finding, ≥ One:**
 - > 250 mg/dL(13.9 mmol/L)
 - < 70 mg/dL(3.9 mmol/L)
 - **Blood glucose monitoring and, ≥ One:** ^(62, 63)
 - Insulin adjustment or dose held ⁽⁶⁴⁾
 - Oral hypoglycemic agent adjustment or dose held ⁽⁶⁵⁾
- ***C. difficile* infection, actual or suspected and, Both:** ^(66, 67, 68)
 - Anti-infective (includes PO) ≤ 5d ^(69, 70)
 - Inadequate oral intake ⁽⁷¹⁾
- **Chest tube and, ≥ One:** ⁽⁷²⁾
 - Suction, continuous
 - Drainage > 2 mL/kg/24h
 - Repositioning within last 24h
 - Water seal within last 24h
- Continued weight loss and initiation or advancement of tube feeding ≤ 3d
- Post Critical care ≤ 24h
- **INTERMEDIATE, ≥ One:**
 - Anti-infective desensitization ≤ 24h ⁽³³⁾
 - **HFNC and, ≥ One:** ^(19, 34)
 - **Flow rate, One:** ⁽²⁰⁾
 - ≤ 20 L/min (> 10 kg)
 - < 2 L/kg/min (≤ 10 kg)
 - FiO₂ > 21%(0.21) to 39%(0.39) ⁽²¹⁾
 - NIPPV ⁽²²⁾
 - Oxygen 35-39%(0.35-0.39) ⁽²¹⁾
- **CRITICAL, ≥ One:**
 - **Acute kidney injury (AKI) and, Both:** ⁽³⁵⁾
 - **Finding, ≥ One:**
 - **Potassium > 6.0 mEq/L(6.0 mmol/L) and, ≥ One:**
 - Loss of P wave
 - Multifocal PVCs
 - Neuromuscular deficit ⁽³⁶⁾
 - Widening QRS
 - **Sodium, One:**
 - < 120 mEq/L(120 mmol/L) and volume overload
 - > 158 mEq/L(158 mmol/L)
 - **Intervention, ≥ One:**
 - Calcium chloride
 - Calcium gluconate
 - Glucose 50%(0.50) with insulin ⁽³⁷⁾
 - Diuretic, continuous
 - IV fluid resuscitation ≤ 2d ⁽²⁹⁾
 - **Dialysis, One:**

- Hemodialysis $\leq$ 7d since initiation
- Peritoneal dialysis $\leq$ 7d since initiation
- CRRT ⁽³⁸⁾
- DIC and blood product transfusion ⁽³⁹⁾
- ECMO or ECLS ^(40, 41)
- Hemodynamic instability, **Both**:
 - Finding, $\geq$ **Two**:
 - Heart rate, **One**: ⁽²⁴⁾
 - $> 180/\text{min}$, sustained (age ≤ 12 mos) ⁽⁸⁾
 - $> 160/\text{min}$, sustained (age > 12 mos to 4 yrs) ⁽⁸⁾
 - $> 150/\text{min}$, sustained (age > 4 yrs to 8 yrs) ⁽⁸⁾
 - $> 140/\text{min}$, sustained (age > 8 yrs to 15 yrs) ⁽⁸⁾
 - $> 130/\text{min}$, sustained (age > 15 yrs to < 18 yrs) ⁽⁸⁾
 - Systolic BP, **One**:
 - < 70 mmHg (age ≤ 12 mos)
 - < 80 mmHg (age > 1 to < 5 yrs)
 - < 90 mmHg (age 5 to < 18 yrs)
 - Labile ⁽²⁵⁾
 - MAP, **One**: ⁽²⁶⁾
 - < 55 mmHg (≤ 12 mos)
 - < 65 mmHg (> 1 to < 18 yrs)
 - Intervention, $\geq$ **One**:
 - Antifibrinolytic ⁽²⁷⁾
 - Blood product transfusion ⁽²⁸⁾
 - IV fluid resuscitation $\leq 2\text{d}$ ⁽²⁹⁾
- Hemodynamic monitoring, invasive ⁽³⁰⁾
- Hemoptysis $> 240\text{ml}/24\text{h}$ and, $\geq$ **One**: ⁽⁴²⁾
 - Lung resection required (**use General Surgical subset**) ⁽⁴³⁾
 - Blood product transfusion ⁽²⁸⁾
 - Bronchial artery embolization performed within last 24h ⁽³¹⁾
 - Post bronchial artery embolization monitoring $\leq 24\text{h}$
- HFNC and, $\geq$ **One**: ^(19, 34)
 - Flow rate, **One**: ⁽³²⁾
 - > 20 L/min (> 10 kg)
 - ≥ 2 L/kg/min (≤ 10 kg)
 - $\text{FiO}_2 \geq 40\%(0.40)$ ⁽²¹⁾
- Mechanical ventilation
- NIPPV and, $\geq$ **One**: ⁽²²⁾
 - $\text{FiO}_2 \geq 40\%(0.40)$
 - Respiratory intervention every 1-2h ⁽¹⁸⁾
- Oxygen $\geq 40\%(0.40)$ ⁽²¹⁾
- Vasopressor, vasodilator, or inotrope continuous (excludes low or fixed dose for end stage heart failure)
- Episode Day 7-9, **One**:
 - ACUTE, **One**:
 - Responder, discharge expected today if clinically stable last 24h, **All**: ⁽⁴⁴⁾
 - Anti-infective regimen, **One**:
 - Complete
 - Established and tolerated, **One**:
 - Transition to PO anti-infective
 - Vascular access established for IV anti-infective
 - O_2 sat $\geq 94\%(0.94)$ or within acceptable limits ⁽⁴⁵⁾
 - Pulmonary function return to baseline or improving ^(7, 46)
 - Tolerating PO or nutritional route established ^(47, 48)

- Treatment needs manageable in post acute setting ⁽⁴⁹⁾
- **Complication or comorbidity, One:**
 - No complication or active comorbidity relevant to this episode of care ⁽⁵⁰⁾
- **Complication or comorbidity and clinically stable for discharge, ≥ One:**
 - **Acute kidney injury (AKI) and, ≥ One:**
 - Kidney function improving or within acceptable limits ⁽⁴⁵⁾
 - Outpatient dialysis regimen established
 - Adverse drug reaction resolving
 - Blood glucose 70-250 mg/dL(3.9-13.9 mmol/L) or within acceptable limits ⁽⁴⁵⁾
 - **C. difficile infection and, Both: ⁽⁶⁸⁾**
 - Decrease in frequency and severity of diarrhea
 - Tolerating PO ⁽⁴⁷⁾
 - Chest tube removed and no reaccumulation of air or fluid by chest x-ray (CXR)
 - Hemoptysis controlled or manageable
- **Partial responder, not clinically stable for discharge and requires continued stay, ≥ One:**
 - **Inadequate clinical improvement and, Both: ⁽⁴⁶⁾**
 - **Finding, ≥ One:**
 - FEV₁ < 80%(0.80) of baseline ^(9, 10)
 - Hemoptysis, persistent
 - O₂ sat < 94%(0.94) and < baseline requiring supplemental oxygen ^(7, 21)
 - Pulmonary infiltrates worsening by imaging
 - Sputum production increasing
 - **Intervention, Both:**
 - **Airway clearance therapy, One: ⁽¹³⁾**
 - At least 3x/24h
 - Contraindicated due to hemoptysis ⁽¹⁴⁾
 - Anti-infective
 - **IV anti-infective regimen incomplete and, Both:**
 - **Barrier to discharge, ≥ One: ⁽⁴⁹⁾**
 - Patient or caregiver unable to adhere to treatment regimen
 - Treatment intensity exceeds resources available in post acute setting
 - **Intervention, Both:**
 - **Airway clearance therapy, One: ⁽¹³⁾**
 - At least 3x/24h
 - Contraindicated due to hemoptysis ⁽¹⁴⁾
 - Anti-infective
 - **AKI, Both: ⁽³⁵⁾**
 - **Finding, ≥ One:**
 - Oliguria ^(51, 52)
 - **Creatinine, One: ⁽⁵³⁾**
 - ≥ 1.5x baseline and chronic kidney disease ⁽⁷⁾
 - ≥ 1.5x ULN
 - GFR > 25%(0.25) decrease from baseline ^(7, 54)
 - Dialysis initiated ⁽⁵⁵⁾
 - **Intervention, ≥ One:**
 - Volume expander ≤ 2d ⁽⁵⁶⁾
 - Diuretic ≥ 2x/24h, ≤ 3d
 - Dialysis (initial) ≤ 7d since initiation ^(55, 57)
 - Dialysis discontinued requiring lab monitoring ≤ 2d since discontinuation ⁽⁵⁸⁾
 - Immunosuppressant therapy ≤ 3d ^(59, 60)
 - Adverse drug reaction requiring change in medication ≤ 24h ⁽⁶¹⁾
 - Anti-infective level non-therapeutic and adjustment in dose or frequency ≤ 2d
 - **Blood glucose ≤ 24h, Both: ⁽⁶²⁾**

- Finding, ≥ **One**:
 - > 250 mg/dL(13.9 mmol/L)
 - < 70 mg/dL(3.9 mmol/L)
- Blood glucose monitoring and, ≥ **One**:^(62, 63)
 - Insulin adjustment or dose held⁽⁶⁴⁾
 - Oral hypoglycemic agent adjustment or dose held⁽⁶⁵⁾
- *C. difficile* infection, actual or suspected and, **Both**:^(66, 67, 68)
 - Anti-infective (includes PO) ≤ 5d^(69, 70)
 - Inadequate oral intake⁽⁷¹⁾
- Chest tube and, ≥ **One**:⁽⁷²⁾
 - Suction, continuous
 - Drainage > 2 mL/kg/24h
 - Repositioning within last 24h
 - Water seal within last 24h
- Continued weight loss and initiation or advancement of tube feeding ≤ 3d
- Post Critical care ≤ 24h
- **INTERMEDIATE**, ≥ **One**:
 - Anti-infective desensitization ≤ 24h⁽³³⁾
- HFNC and, ≥ **One**:^(19, 34)
 - Flow rate, **One**:⁽²⁰⁾
 - ≤ 20 L/min (> 10 kg)
 - < 2 L/kg/min (≤ 10 kg)
 - FiO₂ > 21%(0.21) to 39%(0.39)⁽²¹⁾
- NIPPV⁽²²⁾
- Oxygen 35-39%(0.35-0.39)⁽²¹⁾
- **CRITICAL**, ≥ **One**:
 - Acute kidney injury (AKI) and, **Both**:⁽³⁵⁾
 - Finding, ≥ **One**:
 - Potassium > 6.0 mEq/L(6.0 mmol/L) and, ≥ **One**:
 - Loss of P wave
 - Multifocal PVCs
 - Neuromuscular deficit⁽³⁶⁾
 - Widening QRS
 - Sodium, **One**:
 - < 120 mEq/L(120 mmol/L) and volume overload
 - > 158 mEq/L(158 mmol/L)
 - Intervention, ≥ **One**:
 - Calcium chloride
 - Calcium gluconate
 - Glucose 50%(0.50) with insulin⁽³⁷⁾
 - Diuretic, continuous
 - IV fluid resuscitation ≤ 2d⁽²⁹⁾
 - Dialysis, **One**:
 - Hemodialysis ≤ 7d since initiation
 - Peritoneal dialysis ≤ 7d since initiation
 - CRRT⁽³⁸⁾
 - DIC and blood product transfusion⁽³⁹⁾
 - ECMO or ECLS^(40, 41)
 - Hemodynamic instability, **Both**:
 - Finding, ≥ **Two**:
 - Heart rate, **One**:⁽²⁴⁾
 - > 180/min, sustained (age ≤ 12 mos)⁽⁸⁾
 - > 160/min, sustained (age > 12 mos to 4 yrs)⁽⁸⁾

- > 150/min, sustained (age > 4 yrs to 8 yrs) ⁽⁸⁾
- > 140/min, sustained (age > 8 yrs to 15 yrs) ⁽⁸⁾
- > 130/min, sustained (age > 15 yrs to < 18 yrs) ⁽⁸⁾
- **Systolic BP, One:**
 - < 70 mmHg (age ≤ 12 mos)
 - < 80 mmHg (age > 1 to < 5 yrs)
 - < 90 mmHg (age 5 to < 18 yrs)
 - Labile ⁽²⁵⁾
- **MAP, One:** ⁽²⁶⁾
 - < 55 mmHg (≤ 12 mos)
 - < 65 mmHg (> 1 to < 18 yrs)
- **Intervention, ≥ One:**
 - Antifibrinolytic ⁽²⁷⁾
 - Blood product transfusion ⁽²⁸⁾
 - IV fluid resuscitation ≤ 2d ⁽²⁹⁾
- Hemodynamic monitoring, invasive ⁽³⁰⁾
- **Hemoptysis > 240ml/24h and, ≥ One:** ⁽⁴²⁾
 - Lung resection required (**use General Surgical subset**) ⁽⁴³⁾
 - Blood product transfusion ⁽²⁸⁾
 - Bronchial artery embolization performed within last 24h ⁽³¹⁾
 - Post bronchial artery embolization monitoring ≤ 24h
- **HFNC and, ≥ One:** ^(19, 34)
 - **Flow rate, One:** ⁽³²⁾
 - > 20 L/min (> 10 kg)
 - ≥ 2 L/kg/min (≤ 10 kg)
 - FiO₂ ≥ 40%(0.40) ⁽²¹⁾
- Mechanical ventilation
- **NIPPV and, ≥ One:** ⁽²²⁾
 - FiO₂ ≥ 40%(0.40)
 - Respiratory intervention every 1-2h ⁽¹⁸⁾
- Oxygen ≥ 40%(0.40) ⁽²¹⁾
- Vasopressor, vasodilator, or inotrope continuous (excludes low or fixed dose for end stage heart failure)
- **Episode Day 10, One:**
 - **ACUTE, One:**
 - **Responder, discharge expected today if clinically stable last 24h, All:** ⁽⁴⁴⁾
 - **Anti-infective regimen, One:**
 - Complete
 - **Established and tolerated, One:**
 - Transition to PO anti-infective
 - Vascular access established for IV anti-infective
 - O₂ sat ≥ 94%(0.94) or within acceptable limits ⁽⁴⁵⁾
 - Pulmonary function return to baseline or improving ^(7, 46)
 - Tolerating PO or nutritional route established ^(47, 48)
 - Treatment needs manageable in post acute setting ⁽⁴⁹⁾
 - **Complication or comorbidity, One:**
 - No complication or active comorbidity relevant to this episode of care ⁽⁵⁰⁾
 - **Complication or comorbidity and clinically stable for discharge, ≥ One:**
 - **Acute kidney injury (AKI) and, ≥ One:**
 - Kidney function improving or within acceptable limits ⁽⁴⁵⁾
 - Outpatient dialysis regimen established
 - Adverse drug reaction resolving
 - Blood glucose 70-250 mg/dL(3.9-13.9 mmol/L) or within acceptable limits ⁽⁴⁵⁾
 - **C. difficile infection and, Both:** ⁽⁶⁸⁾

- Decrease in frequency and severity of diarrhea
- Tolerating PO ⁽⁴⁷⁾
- Chest tube removed and no reaccumulation of air or fluid by chest x-ray (CXR)
- Hemoptysis controlled or manageable
- **Non-responder**, not clinically stable for discharge and exceeds episode days at the Acute level of care for this condition, **One**:
 - Use **Extended Stay subset**
 - Refer for secondary review
- **INTERMEDIATE, One**:
 - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Intermediate level of care for this condition, **One**:
 - Use **Extended Stay subset**
 - Refer for secondary review
- **CRITICAL, One**:
 - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Critical level of care for this condition, **One**:
 - Use **Extended Stay subset**
 - Refer for secondary review

Discharge, Level of Care, One: ⁽⁷³⁾

- **HOME, Both**:
 - Level of care appropriateness, **Both**:
 - Home environment safe and accessible ⁽⁷⁴⁾
 - Patient or caregiver demonstrates ability to manage care
 - Cystic fibrosis discharge planning, **All**:
 - Provide comprehensive written discharge and teaching instructions ⁽⁷⁵⁾
 - Education on the importance of, **All**:
 - Compliance with maintenance care
 - Infection prevention ⁽⁷⁶⁾
 - Recognizing the signs and symptoms of pulmonary exacerbation
 - Medication reconciliation ⁽⁷⁷⁾
 - Follow-up care planned ⁽⁷⁸⁾
 - Identify and address transportation needs
- **HOME CARE, Both**:
 - Level of care appropriateness, **All**:
 - Home environment safe and accessible ⁽⁷⁴⁾
 - Patient or caregiver willing and able to learn care
 - Outpatient management contraindicated or unavailable ^(79, 80)
 - Skilled services, ≥ **One**: ⁽⁸¹⁾
 - Skilled nursing ⁽⁸²⁾
 - Skilled therapy, PT, OT, or ST
 - Behavioral health ⁽⁸³⁾
 - Cystic fibrosis discharge planning, **All**:
 - Provide comprehensive written discharge and teaching instructions ⁽⁷⁵⁾
 - Education on the importance of, **All**:
 - Recognizing the signs and symptoms of pulmonary exacerbation
 - Compliance with maintenance care
 - Infection prevention ⁽⁷⁶⁾
 - Medication reconciliation ⁽⁷⁷⁾
 - Follow-up care planned ⁽⁷⁸⁾
 - Identify and address transportation needs
- **BEHAVIORAL HEALTH, Both**:
 - Level of care appropriateness, **One**:

- Support systems available ⁽⁸⁴⁾
- **Skilled treatment, ≥ One:**
 - Clinical assessment ⁽⁸⁵⁾
 - Individual, group, or family counseling
 - Psychiatric, substance, or medication evaluation
- **SKILLED MEDICAL OR THERAPY, Both:**
 - **Level of care appropriateness, All:**
 - Medical practitioner, NP, PA assessment or oversight ≥ 1x/wk
 - Treatment precluded in a lower level
 - **Services, ≥ One:**
 - Medical and skilled nursing interventions or assessment 1-2x/24h ⁽⁸²⁾
 - **Therapy, All:**
 - Goal directed therapy with rehabilitation potential ^(86, 87)
 - Functional impairments requiring at least supervision ⁽⁸⁸⁾
 - Able to tolerate 1h/d of skilled therapy ≥ 5d/wk
 - **Complete prior to facility transfer, All:**
 - Medication reconciliation ⁽⁷⁷⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **SUBACUTE MEDICAL OR THERAPY, Both:**
 - **Level of care appropriateness, All:**
 - Medical practitioner, NP, PA assessment or oversight ≥ 2x/wk
 - Treatment precluded in a lower level
 - **Services, ≥ One:**
 - Medical and skilled nursing interventions or assessment 4h/24h ⁽⁸²⁾
 - **Therapy, All:**
 - Goal directed therapy with rehabilitation potential ^(86, 87)
 - ≥ 2 functional impairments requiring at least minimum assistance ⁽⁸⁸⁾
 - Able to tolerate 2-3h/d of skilled therapy ≥ 5d/wk and ≥ 1 therapy discipline
 - **Complete prior to facility transfer, All:**
 - Medication reconciliation ⁽⁷⁷⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **SUBACUTE REHAB, Both:**
 - **Level of care appropriateness, All:**
 - Medical practitioner, NP, PA assessment or oversight ≥ 3x/wk
 - Treatment precluded in a lower level
 - **Services, All:**
 - Goal directed therapy with rehabilitation potential ^(86, 87)
 - ≥ 2 functional impairments requiring at least minimum assistance ⁽⁸⁸⁾
 - Able to tolerate 2-3h/d of skilled therapy ≥ 5d/wk and ≥ 2 therapy disciplines
 - **Complete prior to facility transfer, All:**
 - Medication reconciliation ⁽⁷⁷⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **ACUTE REHAB, Both:**
 - **Level of care appropriateness, Both:**
 - Medical practitioner, NP, PA assessment or oversight ≥ 3x/wk
 - **Services, All:**
 - Goal directed therapy with rehabilitation potential ^(86, 87)

- ≥ 2 functional impairments requiring at least minimum assistance ⁽⁸⁸⁾
 - Able to tolerate ≥ 15 h/wk of skilled therapy and ≥ 2 therapy disciplines ⁽⁸⁹⁾
- Complete prior to facility transfer, **All:**
 - Medication reconciliation ⁽⁷⁷⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **LONG TERM ACUTE CARE, Both:**
 - Level of care appropriateness, **Both:**
 - Treatment precluded in a lower level
 - In lieu of acute or continued hospitalization or failed lower level of care, $\geq$ **One:**
 - Primary condition or illness and treatment of active comorbid condition(s) ⁽⁹⁰⁾
 - Ventilator dependent ≥ 6 h/d and weaning planned ^(91, 92)
 - Acute and comorbid conditions requiring prolonged hospitalization ⁽⁹³⁾
- Complete prior to facility transfer, **All:**
 - Medication reconciliation ⁽⁷⁷⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation

Notes:**1:**

Hemodynamically stable patients who require treatment, assessment, or intervention every 4 to 8 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions are based on a variety of factors including medical necessity, payer rules, and contracts.

2:

The precise definition of a pulmonary exacerbation remains unclear, but is characterized by a combination of clinical signs and symptoms indicating respiratory decline including: changes in cough, sputum production, work of breathing, appetite, weight, pulmonary function tests, oxygenation, and imaging findings (Plummer et al., Cochrane Database Syst Rev 2016, 9: Cd006682).

3:

Expected treatments for a pulmonary exacerbation of cystic fibrosis include: increasing the frequency of airway clearance, treatment with inhaled bronchodilators, chest physical therapy and postural drainage, use of antibiotics, and increasing the dose of mucolytic agents (Plummer et al., Cochrane Database Syst Rev 2016, 9: Cd006682).

4:

Failed treatment is defined as, at least 1 week on a new anti-infective regimen accompanied by an increased frequency or dose of nebulizer, without evidence of clinical improvement.

5:

Hemoptysis is a common symptom in patients with cystic fibrosis. It can occur during an acute pulmonary exacerbation (most commonly with *P. aeruginosa* positive sputum culture) and can range from scant to life-threatening amounts. In most cases, hemoptysis will resolve spontaneously or during treatment of the respiratory exacerbation. Interventions to control bleeding, such as bronchial artery embolization, are reserved for unstable patients with massive hemoptysis (West et al., J Cyst Fibros 2017:).

6:

Hypoxia in the setting of a pulmonary exacerbation may be defined by a new or increased supplemental oxygen requirement, or a decline in room air oxygen saturation greater than 4% from baseline (Zemanick et al., Pediatric pulmonology 2010, 45: 569-77).

7:

Baseline refers to either the patient's normal baseline or a newly established baseline. In the absence of documentation, a patient's baseline status may be presumed to be normal.

8:

Vital signs are considered sustained findings when they are abnormal for 2 or more readings greater than 15 minutes apart or are lasting at least 15 minutes. This excludes an isolated reading, a transient abnormal measurement, or a finding that requires urgent treatment (e.g., severe hypoxemia, ventricular arrhythmia, hypotension). Application of criteria for a vital sign finding should be based on confirmatory measurements repeated at regular intervals. Events that are infrequent or vital sign abnormalities that have resolved with outpatient treatment are not considered to be sustained (e.g., resolution of hypertension following anti-hypertensive therapy, resolution of tachycardia following rehydration).

Note: The definition of sustained reflects the opinion of InterQual's® external peer reviewers. This criteria point is based upon current best practice and is the product of an iterative process involving multiple clinicians with diverse expertise in varied geographic settings.

9:

Baseline forced expiratory volume over one second (FEV₁) is the standard method used to establish the progression of disease in a patient with cystic fibrosis. The baseline FEV₁ refers to the patient's documented value (within 3

months) prior to the development of an acute pulmonary exacerbation(Plummer et al., Cochrane Database Syst Rev 2016, 9: Cd006682).

10:

In an effort to address inequities in medicine and improve appropriate access to care, multiple medical societies have been reevaluating the use of adjustments made for a patient's race and updating their calculators and reference equations when reporting diagnostic results.

In 2023, the American Thoracic Society issued updated guidance on reporting of pulmonary function tests (PFTs), advocating for a race-neutral average reference equation to determine the baseline (i.e., predicted normal) for a given patient (Bhakta et al., Am J Respir Crit Care Med 2023, 207: 978-995). The reference equation previously included lower predicted normal values for patients of certain races and ethnicities (in the US, for Black and Asian patients); a practice which has not been found to have biological validity and may lead to misclassification of diagnoses and disease severity in racial or ethnic minorities (Vyas et al., N Engl J Med 2020, 383: 874-882). Since not all spirometer manufacturers and PFT laboratories have switched over to the new race-neutral average reference equation, clinical judgement and consideration of patient-specific factors should be used when interpreting PFT results and deciding whether patients with borderline reported PFT values in the presence of concerning symptoms or signs should have further evaluation or treatment.

11:

The Centers for Disease Control and Prevention (CDC) define a fever as greater than or equal to 100.4 degrees Fahrenheit (i.e., greater than or equal to 38 degrees Celsius) and do not specify the route of measurement (Centers for Disease Control and Prevention, Definitions of symptoms for reportable illness. 2017). The route utilized may be directed by institutional protocols, equipment availability, patient preference, or other patient-specific factors.

12:

During a pulmonary exacerbation, patients should receive a nutrition consult to ensure optimization of nutrition, which is essential to meeting increased energy demands and promoting recovery. Nutrition management may include consuming an unrestricted diet high in calories and protein, pancreatic enzyme replacement therapy, and supplementation. Patients with cystic fibrosis-related diabetes should follow the same dietary guidelines described above. In order to achieve glycemic control, carbohydrate counting and use of insulin-to-carbohydrate ratios are recommended (Moran et al., Diabetes Care 2010, 33: 2697-708).

13:

Airway clearance therapy (ACT) is used to optimize pulmonary function by clearing sputum from the respiratory tract. The most effective techniques are:

- Chest physiotherapy (CPT): Percussion and vibration of the chest and can be combined with modified postural drainage (body positioning without the head down) to remove mucus from the lungs.
- Positive expiratory pressure (PEP): A mask or mouthpiece and breathing against applied pressure (10-20 cm H₂O). Oscillating PEP (not recommended in children), vibrates the airflow during PEP
- Active-cycle-of-breathing technique (ABCT): A cycle of deep controlled breathing, and forced thoracic expiration to clear secretions (used in all stages of disease)
- Autogenic drainage (AD): A method of controlled breathing to move mucus from the small airways to the larger airways for expulsion
- High-frequency chest compression (use of a compressive vest to vibrate the airways)

There is insufficient evidence to recommend one technique over another, and treatment should be individualized according to age, patient preference, and risk for adverse effects. The Cystic Fibrosis Foundation recommends that airway clearance therapy increase in duration or frequency during the treatment of pulmonary exacerbations (Strickland et al., Respir Care 2015, 60: 1071-7).

14:

Contraindicated refers to **documented** clinical circumstances when a medication or treatment is not indicated based on the nature of the disease, comorbid condition, or a drug allergy or intolerance.

15:

Hemodynamically stable patients who require treatment, assessment, or intervention every 2 to 4 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions are based on a variety of factors including medical necessity, payer rules, and contracts.

16:

For patients at the Acute level of care, mental status change may include disorientation, agitation, increasing irritability, listlessness, or lethargy. The inability to soothe an infant, abnormal cry (e.g., weak, high-pitched), or decreased social interaction with parents or caregivers in children 2 years old or younger, may also qualify as a mental status change (Keith Kleinman, In: Harriet Lane Handbook, 22nd edn. 2021).

Instruction: Patients with acute coma, stupor, or somnolence should be reviewed at a higher level of care due to the need for more frequent neurological monitoring. These criteria exclude chronic coma or stupor.

17:

Instruction: Room air assessment may not be clinically appropriate in patients who are dyspneic, tachypneic, cyanotic, or experiencing moderate to severe respiratory distress. If a patient remains hypoxic while on oxygen, room air assessment is not required.

18:

Respiratory interventions include ventilator/noninvasive positive pressure ventilation (NIPPV) setting changes, chest physiotherapy, suctioning, and nebulizer or metered-dose inhaler (MDI) treatments.

19:

High flow nasal cannula (HFNC) delivers a fraction of inspired oxygen (FiO_2) that is heated (up to 37 degrees Celsius) with 100%(1.0) relative humidity at flow rates set independent from the FiO_2 . The heated and humidified gas has been shown to improve comfort, aid in the clearance of secretions, and reduce bronchospasms. When flow rates are set above the patient's inspiratory demand, the upper airway is washed out of carbon dioxide, nasal resistance is decreased, and there is a reduction in dead space, leading to improved work of breathing. In addition, HFNC reduces the amount of ambient air a patient can breathe in, resulting in less oxygen dilution. Bronchiolitis is the main indication for HFNC in pediatric patients beyond the neonatal period. Although the benefits of HFNC in other conditions such as pneumonia, asthma, and post-extubation have been reported, more well-designed studies are needed to guide the use of HFNC in these conditions (Kwon, Clin Exp Pediatr 2020, 63: 3-7).

20:

High flow nasal cannula (HFNC) consists of three parameters: temperature, fraction of inspired oxygen (FiO_2), and flow rate. Each parameter is independent of the other. The flow rate is how fast the heated and humidified air/oxygen is delivered per minute and is documented as liters per minute (L/min). Standard HFNC flow rates in pediatrics have yet to be established (Kwon, Clin Exp Pediatr 2020, 63: 3-7). Although most studies on HFNC in pediatrics focus on children with bronchiolitis, they can provide some guidance on appropriate flow rates outside of the intensive care unit (ICU). Initial flow rates in clinical trials have found that 1-2 L/kg/min is effective in patients with bronchiolitis (Yurtseven et al., Pediatr Pulmonol 2019, 54: 894-900). However, a recent survey revealed that only 19%(0.19) of responders use a weight-based flow rate method (Miller et al., Respir Care 2018, 63: 894-9). This method may not be optimal in older children as it can result in very high flow rates that may not be tolerated. Some studies combat this potential problem by setting maximum flow rates. Flow rates up to 20 L/min are well tolerated on non-ICU wards (Willer et al., Hosp Pediatr 2021, 11: 891-5; Kepreotes et al., Lancet 2017, 389: 930-9). Therefore, a flow rate of 20 L/min was chosen as the maximum for the Intermediate level of care.

Instruction: To calculate the L/kg/min, the flow rate is divided by the patient's weight.

Example:

HFNC flow rate: 12 L/min

Patient weight: 8 kg

$12/8 = 1.5$

This patient is on a flow rate of 1.5 L/kg/min

21:

Oxygen therapy is the administration of oxygen at concentrations greater than ambient air [room air: 21%(0.21) FIO₂] to correct hypoxia. For pediatric patients, the selection of an oxygen delivery system needs to take into consideration the patient's size, needs, and therapeutic goals. The oxygen concentration or percentage (FIO₂) delivered varies with the manufacturer's design, delivery device, patient's minute ventilation (dependent on the respiratory rate and tidal volume), and oxygen flow rate. In general, patients with higher minute ventilation (common in most sick patients and those with respiratory disease) require considerably higher flow rates to correct hypoxia. The following are examples of oxygen delivery systems commonly used for pediatric patients with the percentage range of FIO₂ optimally delivered:

Low-flow systems

- Nasal cannula / Nasopharyngeal catheters 24-35%(0.24-0.35) FIO₂

Reservoir systems (For neonates, infants, and young children, the maximum FIO₂ achievable is not well documented)

- Simple oxygen masks 35-50%(0.35-0.50) FIO₂
- Partial-rebreathing masks 40-60%(0.40-0.60) FIO₂
- Non-rebreathing masks > 60%(0.60) FIO₂

High-flow systems

- Air-entrainment masks / nebulizers 24-40%(0.24-0.40) FIO₂
- Heated, humidified, high flow oxygen concentrating nasal cannula 21-100%(0.21-1.0) FIO₂

Enclosure systems

- Oxygen hoods / tents 21-100%(0.21-1.0) FIO₂

For children less than 5 years old, the oxygen regimen may need to be adjusted to a lesser concentration and flow rate, while still requiring acute level of care due to the patient's size, metabolic needs, and vital sign instability.

22:

Noninvasive positive pressure ventilation (NIPPV), also known as NIV, provides respiratory support by application of a tightly fitting facial mask, nasal mask, or helmet rather than an endotracheal tube or tracheostomy. In some cases, the use of NIPPV can avoid the need for endotracheal intubation and decreases the risk of barotrauma, lung injury, and/or infection. NIPPV is commonly delivered by a bilevel positive airway pressure ventilator (BiPAP), a continuous positive airway pressure device (CPAP), or a mechanical ventilator. Supplemental oxygen can be delivered at concentrations approximating 100%(1.0).

The decision to provide respiratory support via NIPPV, and the modality to provide NIPPV, is based upon the patient's specific clinical findings (e.g., medical condition leading to respiratory failure, underlying comorbidities, clinical progression, improvement). Examples of NIPPV devices are volumetric (i.e., deliver a defined volume), barometric (i.e., deliver a defined pressure), and combined (i.e., deliver defined volumes and pressure). The terminology for NIPPV delivery systems may differ between equipment manufacturers and provider organizations. InterQual® criteria do not differentiate between the different NIPPV modalities.

Whether or not high-flow nasal cannula (HFNC) should be classified as a component of NIPPV is unclear, and there is conflicting evidence in the medical literature. Where HFNC is an appropriate modality, InterQual® defines this in a specific criteria point. As such, NIPPV does not include HFNC (Hackett, A., PulmCCM 2018; Nardi et al., F1000Res 2017, 6: 290; Osadnik et al., Cochrane Database Syst Rev 2017, 7: CD004104; Allison and Winters, Emerg Med Clin North Am 2016, 34: 51-62; Gregoretti et al., Crit Care Clin 2015, 31: 435-57).

23:

Hemodynamically unstable patients (or those with the potential to become unstable) who require treatment, assessment, or intervention every 1 to 2 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions are based on a variety of factors including medical necessity, payer rules, and contracts.

24:

In the pediatric patient, hemodynamic instability could be assessed by an elevated heart rate (HR) above the 95th percentile norms based on specific age groups. A combination of different parameters and their trends could also be utilized to ascertain impending clinical deterioration more accurately in these patients to allow for closer

monitoring in the critical care setting, including respiratory rate, mental status, and capillary refill time, amongst others. Other factors that could impact HR in the pediatric population include pain, anxiety, medications, and/or fever. Although no specific consensus has been reached on HR thresholds that signify unstable hemodynamics by various pediatric societies, systemic reviews and observational studies have compared HR data to textbook references such as the Pediatric Advance Life Support (PALS). Eytan et al., in 2017, studied HR in pediatric patients who were critically ill in the pediatric intensive care unit (PICU) and developed percentiles of heart rates categorized into the same age groups used in previous studies, including PALS, with the goal of comparing their observed norms, especially between the 5th and 95th percentiles (Eytan et al., *Frontiers in Pediatrics* 2017, 5). Using their HR data above the 95th percentile, HR thresholds were utilized to identify the pediatric patient who may require immediate or urgent intervention and rapidly treat unstable hemodynamics to prevent a potentially serious event.

25:

Blood pressure is considered labile when it is extremely variable and difficult to control.

26:

In pediatric patients, hemodynamic instability may be defined as a mean arterial pressure (MAP) less than 55 mmHg in infants up to 1 year of age and less than 65 mmHg in patients over 1 year of age (Matics and Sanchez-Pinto, *JAMA Pediatr* 2017, 171: e172352).

27:

Antifibrinolytic medications reduce blood loss by inhibiting clot degradation. Specific agents include tranexamic acid and aminocaproic acid, and may be indicated in bleeding due to traumatic, surgical, obstetric, gynecological, or gastrointestinal bleeding. Efficacy for these indications is in various stages of investigation, with the strongest evidence supporting use in bleeding due to trauma and surgery (Steinmetzer et al., *J Med Chem* 2020, 63: 1445-72; Draxler and Medcalf, *Transfus Med Rev* 2015, 29: 102-9; Hunt, *Anaesthesia* 2015, 70 Suppl 1: 50-3, e18).

28:

The volume of blood product infused in a pediatric patient is dependent on the weight of the child. For some children, small volume transfusions of blood products (especially packed red blood cells) are necessary to prevent fluid overload. An appropriate transfusion volume would be 10 to 15 mL/kg, infused at a rate of 2 to 4 mL/kg per hour (Park, In: Elsevier, ed. *The Harriet Lane Handbook*, 23 edn. 2024:365-407; Rodrigue and Sloan, In: *Hematology*, 8 edn. 2023:1892-9).

29:

Fluid resuscitation involves the administration of repeated intravenous fluid boluses to acutely restore depleted intravascular volume. There is no evidence to support the use of colloids over crystalloids; studies have demonstrated that the primary choice for a resuscitation fluid is a balanced crystalloid solution, such as Lactated Ringer's (Myburgh, *N Engl J Med* 2018, 378: 862-3).

Bolus volumes for administration may be generally defined by the following:

- Fluid boluses of 250 to 1,000 milliliters in adults (Scales, *Br J Nurs* 2014, 23; Myburgh and Mythen, *N Engl J Med* 2013, 369: 1243-51; Vincent and De Backer, *N Engl J Med* 2013, 369: 1726-34)
- Fluid boluses of 10 to 20 milliliters per kilogram in pediatric patients. In some cases, boluses of 60 milliliters per hour or more may have to be administered in the first hour (Davis et al., *Crit Care Med* 2017, 45: 1061-93)

For patients with decreased cardiac reserve, significant renal dysfunction, or third-spacing, smaller volume boluses may be used (National Clinical Guideline Centre for Acute and Chronic Conditions, *IV Fluids in Children: Intravenous Fluid Therapy in Children and Young People in Hospital*. 2015. Reaffirmed 2020). The total number of fluid boluses required to restore intravascular volume and avoid fluid overload varies by patient (Joseph et al., *J Trauma Acute Care Surg* 2014, 76: 457-61; Holodinsky et al., *Crit Care* 2013, 17: R249; Kirkpatrick et al., *Intensive Care Med* 2013, 39: 1190-206). Dynamic measures such as bedside vena cava ultrasound and monitoring of urine output may assist in determining when volume resuscitation has been achieved (Kent et al., *J Surg Res* 2013, 184: 561-6; Zeng et al., *Am J Emerg Med* 2013, 31: 763-7; Weekes et al., *Acad Emerg Med* 2011, 18: 912-21).

30:

Invasive hemodynamic monitoring may include at least one of the following methods of assessment: arterial line, pulmonary artery catheter, central venous pressure, or cardiac output monitoring (Rajaram et al., Cochrane Database Syst Rev 2013, 2: CD003408).

31:

Bronchial artery embolization should be performed in cystic fibrosis patients with massive hemoptysis who are clinically unstable. The aim of this treatment is to embolize the bleeding vessel in an effort to stop the bleeding (Flume et al., American journal of respiratory and critical care medicine 2010, 182: 298-306).

32:

High flow nasal cannula (HFNC) consists of the following three parameters: temperature, fraction of inspired oxygen (FiO_2), and flow rate. Each parameter is independent of each other. The flow rate is how fast the heated and humidified air/oxygen is delivered per minute and is documented as liters per minute (L/min). The flow rate is often based on the patient's weight in smaller children (Kwon, Clin Exp Pediatr 2020, 63: 3-7). To calculate the liters per kilogram per minute (L/kg/min), the flow rate is divided by the patient's weight.

Example:

HFNC flow rate: 12 L/min

Patient weight: 8 kg

$12/8 = 1.5$

This patient is on a flow rate of 1.5 L/kg/min

33:

Desensitization is the safest measure of reintroducing a drug when alternative medicines are not tolerated. Due to frequent anti-infective exposure, cystic fibrosis patients often develop allergic reactions to multiple agents. Desensitization protocols involve the administration of increasing concentrations and doses of an anti-infective medication in frequent intervals. Patients are monitored closely for anaphylaxis and bronchospasm (Whitaker et al., J Cyst Fibros 2011, 10: 282-5; Legere et al., Journal of cystic fibrosis: official journal of the European Cystic Fibrosis Society 2009, 8: 418-24).

34:

Currently, there is no universally accepted method for weaning pediatric patients off high-flow nasal cannula (HFNC), and weaning protocols vary based on institutional policies. However, one commonly used method is to gradually decrease the fraction of inspired oxygen (FiO_2) to a specific level, such as 40%(0.40), then gradually reduce the flow rate before transitioning to a low-flow oxygen delivery system (e.g., nasal cannula). Less common methods include weaning the FiO_2 to a predetermined value, then transitioning to a low-flow oxygen system without reducing the flow rate, or gradually reducing the flow rate while keeping the FiO_2 constant (Kawaguchi et al., Pediatr Crit Care Med 2020, 21: e228-e35). There is no standard flow rate at which HFNC should be discontinued, but based on the literature, it is suggested to discontinue HFNC when the flow rate reaches 2 L/min in infants, and 4 L/min in children older than 12 months (Udurgucu et al., Pediatr Allergy Immunol Pulmonol 2022, 35: 79-85; Peterson et al., Respir Care 2021, 66: 591-9).

Instruction: When weaning a patient off high-flow nasal cannula (HFNC), it is expected that the current level of care be maintained or a lower level of care from the previous level be utilized to conduct the review if HFNC criteria are used. A typical weaning strategy usually does not involve simultaneously decreasing the flow rate while increasing the FiO_2 . This scenario may indicate that a patient is being kept on the HFNC machine but may be able to tolerate settings that could be administered by nasal cannula. For instance, a patient who was previously on HFNC 10 L/min, 30%(0.30) FiO_2 might be weaned to 3 L/min, but FiO_2 is increased to 100%(1.0). In this case, the patient should be assessed based on the flow rate rather than the FiO_2 .

35:

Acute kidney injury (AKI) is a term used to describe the broad spectrum of kidney function impairment, including insufficiency, oliguria, and failure. The spectrum ranges from minor changes in renal function markers (e.g., increase in serum creatinine, decreased urine output) to failure requiring renal replacement therapy.

AKI is defined as a sudden decrease in renal function resulting from multiple etiologies, including intrinsic kidney disease, ischemia, nephrotoxicity, and extrarenal pathology.

AKI encompasses patients with chronic kidney disease who experience an acute deterioration. AKI is associated

with significant morbidity and mortality in hospitalized patients (Ostermann et al., Kidney Int 2020, 98: 294-309; Guideline Updates, In: Acute kidney injury: prevention, detection and management. 2019; Kidney Disease: Improving Global Outcomes (KDIGO), Kidney International Supplements 2012, 2: ii.-138; Wang et al., Am J Nephrol 2012, 35: 349-55).

36:

Hyperkalemia may be associated with weakness leading to muscle twitching, paresthesia, and paralysis.

37:

Dextrose 50%(0.50) with insulin is used for the treatment of hyperkalemia. This treatment shifts potassium intracellularly, and repeated doses can be given if the hyperkalemia persists. Other treatments that may be used simultaneously include potassium binding agents, diuretics, nebulized albuterol, calcium, sodium bicarbonate, and dialysis (Depret et al., Ann Intensive Care 2019, 9: 32).

38:

While no randomized, controlled trials have shown that continuous renal replacement therapy (CRRT) is superior to intermittent dialysis with respect to survival, there are several advantages that may influence its use despite the lack of demonstrated survival benefit.

These advantages include:

- Gradual solute and fluid shifts resulting in greater hemodynamic stability, little risk of disequilibrium syndrome, and no worsening of brain edema
- Less need for fluid or nutritional restrictions, allowing for adequate nutrition without compromising fluid balance
- Decreased risk for intradialytic hypotension and recurrent kidney injury

Comprehensive guidelines recommend the use of CRRT for patients with hemodynamic instability, brain edema, persistent metabolic acidosis, and large fluid removal requirements. CRRT can be discontinued once renal recovery has been confirmed or another form of renal replacement is chosen based on the patient's clinical condition (Ostermann et al., Kidney Int 2020, 98: 294-309; Depret et al., Ann Intensive Care 2019, 9: 32; Kidney Disease: Improving Global Outcomes (KDIGO), Kidney International Supplements 2012, 2: ii.-138).

39:

Disseminated intravascular coagulation or DIC is a hematological emergency where there is abrupt systemic cascade activation of coagulation leading to thrombosis of midsize to small blood vessels rapidly leading to acute organ dysfunction. Simultaneously, there is consumption of platelets and proteins involved in the coagulation cascade, resulting in acute thrombocytopenia and low clotting factors. DIC can be caused by conditions such as sepsis, trauma, severe obstetrical complications, and malignancies. Laboratory abnormalities found in DIC include low fibrinogen (not always present), elevated fibrin degradation products, elevated D-dimer, thrombocytopenia, and prolonged prothrombin (PT) and activated partial thromboplastin times. Prompt therapy for DIC includes treatment against the underlying disease but may also include platelet transfusion, plasma (fresh frozen plasma; FFP) or factor transfusion in cases of hemorrhage, or anticoagulants in situations with thrombosis formation predominates (Wada et al., J Thromb Haemost 2013:).

40:

Transition Plan: InterQual's® Transition Plan identifies patients at high risk for readmission who may benefit from a comprehensive discharge plan; 60%(0.60) of patients who are extracorporeal membrane oxygenation (ECMO) survivors are readmitted (Jen and Shew, Pediatrics 2010, 125: 1217-23).

41:

Extracorporeal membrane oxygenation (ECMO) or extracorporeal life support (ECLS) provides oxygen and removes carbon dioxide from blood. ECMO/ECLS facilitates the recovery of cardiopulmonary function when other measures fail to improve gas exchange. Indications for the use of ECMO/ECLS in children include refractory septic or cardiogenic shock, cardiac arrest as a bridge to transplant, or refractory hypoxemic or hypercarbic respiratory failure. There is clear evidence to support the use of ECMO/ECLS in the pediatric population, but definitive studies in adults are lacking. However, in adults, observational studies and other data support the use of ECMO/ECLS for

severe refractory, hypoxemic, or hypercarbic respiratory failure (Mehta et al., American College of Cardiology Perspectives and Analysis 2018; Napp et al., Herz 2017, 42: 27-44; Guirand et al., J Trauma Acute Care Surg 2014, 76: 1275-81). ECMO/ECLS incorporates the use of an external circuit, a cannula for vascular access, a pump to propel the blood through the circuit, and an artificial membrane that oxygenates the blood. Common complications include hemorrhage, hemolysis, and infection. Stroke, embolism, and seizures in children have also been reported. Infants with intraventricular hemorrhage (IVH) or those at high risk for IVH (i.e., weight less than 2000 grams, age less than 35 weeks) are not candidates for this therapy (Abrams et al., J Am Coll Cardiol 2014, 63: 2769-78; Domico et al., Pediatr Crit Care Med 2012, 13: 16-21; MacLaren et al., Intensive Care Med 2012, 38: 210-20; Peek et al., Health Technol Assess 2010, 14: 1-46; Mugford et al., Cochrane Database Syst Rev 2008: CD001340).

42:

Massive hemoptysis in Cystic Fibrosis is defined as, a quantity of greater than 240 ml over 24 hours, blood loss that is life threatening, or repeated large amounts of bleeding over a short period of time (days) (Hurt and Simmonds, Paediatr Respir Rev 2012, 13: 200-5).

43:

In the treatment of cystic fibrosis patients with massive hemoptysis, lung resection should be performed as a last resort after all other measures have failed (e.g., bronchial artery embolization) (Flume et al., American journal of respiratory and critical care medicine 2010, 182: 298-306).

44:

Selection of this criteria point indicates that the patient is responding to treatment and is clinically stable for transfer or discharge. To determine the most appropriate post-acute level of care, see discharge criteria.

45:

When a criteria point states "within acceptable limits," it refers to either the patient's normal baseline, a newly established baseline, or parameters that the medical practitioner determines are acceptable.

46:

Patients admitted with pulmonary exacerbations may not return to prior respiratory function with treatment. At risk patients for permanent pulmonary decline include those with advanced disease and multiple prior exacerbation episodes (Cogen et al., Pediatrics 2017, 139:).

47:

Oral intake should be greater than or equal to the patient's maintenance fluid requirements or within parameters which the medical practitioner determines are acceptable. Maintenance fluid requirements should be increased in the presence of dehydration or insensible loss. Maintenance fluid requirements (*mL/kg/d*) can be calculated based on current body weight (*kg*):

- 100 *mL/kg/d* for first 10 *kg*
- 50 *mL/kg/d* for second 10 *kg*
- 20 *mL/kg/d* for each additional *kg* in weight

48:

Routes for nutrition may be by mouth (PO), intravenously (IV), jejunostomy tube (J-tube), or gastrostomy tube (G-tube).

49:

Since treatment of a pulmonary exacerbation involves a combination of aggressive respiratory care and intravenous anti-infective therapy, continued hospitalization is recommended if the resources and support required to treat the exacerbation exceed that which can be provided in the post acute setting (Cogen et al., Pediatrics 2017, 139:). Barriers to discharge may include the following:

- Patient and/or caregiver skill or compliance with the treatment regimen in question
- Medication administration and airway clearance therapy cannot be maintained in the post acute setting at the prescribed frequency or intensity

50:

Relevant complications or comorbidities are conditions that actively require management during an inpatient stay.

51:

Instruction: Measuring urine output in newborns, infants, or patients who are unable to use a bedpan or other measuring device can be done via straight catheterization, or alternatively, by weighing diapers. Weight in grams can be translated to milliliters (1 gram = 1 mL) (Fuhrman, *Pediatr Crit Care Med* 2019, 20: 381-2).

52:

Urine output less than 1 mL/kg/h or 180 mL/m²/24h is commonly considered oliguria in the pediatric patient.

53:

Stage II acute kidney injury includes creatinine greater than 2 times the baseline with chronic disease or upper limit of normal (Ostermann et al., *Kidney Int* 2020, 98: 294-309; *Kidney Disease: Improving Global Outcomes (KDIGO)*, *Kidney International Supplements* 2012, 2: ii.-138).

54:

Glomerular filtration rate (GFR), serum creatinine (sCr), and urine output are all useful indicators of kidney function. While a decrease in GFR is believed to be the most precise estimate of renal impairment, it is difficult to measure and is usually estimated from sCr. Thus, increase in sCr or decrease in urine output are used to denote a potential acute kidney injury (*Kidney Disease: Improving Global Outcomes (KDIGO)*, *Kidney International Supplements* 2012, 2: ii.-138).

55:

Peritoneal dialysis is used more commonly in young children and infants, whereas hemodialysis is preferred for juveniles and teenagers (Chand et al., *Am J Kidney Dis* 2017, 69: 278-86).

Some advantages of peritoneal dialysis over hemodialysis for young children include (Barhight et al., *World J Pediatr Congenit Heart Surg* 2018, 9: 696-704):

- It is simpler to apply
- Anticoagulation not required
- No need for large bore venous access

56:

For expansion of intravascular volume in patients with acute kidney injury (AKI), guidelines recommend initial treatment with an isotonic crystalloid solution (sodium chloride or sodium bicarbonate); the effects of sodium chloride on renal outcomes are being studied. Forced diuresis is not recommended because it can worsen AKI in volume-responsive patients (Ostermann et al., *Kidney Int* 2020, 98: 294-309; *Guideline Updates, In: Acute kidney injury: prevention, detection and management*. 2019; Pistolesi et al., *J Nephrol* 2018, 31: 797-812; Joannidis et al., *Intensive Care Med* 2017, 43: 730-49; *Kidney Disease: Improving Global Outcomes (KDIGO)*, *Kidney International Supplements* 2012, 2: ii.-138).

57:

Initial course refers to the initiation of a chronic hemodialysis/peritoneal dialysis program or initiation for acute renal pathology. A repeat initiation for a recurrence or exacerbation of an acute renal problem, provided that there was a gap of more than 14 days between treatments, would also qualify as an initial course.

58:

Following the discontinuation of dialysis, lab monitoring is necessary for continued management and may include blood urea nitrogen (BUN), creatinine, and electrolytes.

59:

Immunosuppressant medications are used to treat conditions such as graft-versus-host disease (GVHD), hemolytic anemia, organ transplant, graft failure, rejection, inflammatory bowel disease, acute glomerulonephritis, and

inflammatory cellulitis. Medications used for immunosuppression include prednisone, prednisolone, cyclosporine, azathioprine, mycophenolate mofetil, tacrolimus, sirolimus, everolimus, belatacept, antithymocyte globulin (ATG), and basiliximab.

60:

Instruction: If a patient is clinically stable, has no other medical reason for an acute level of care, and has tolerated 24 to 48 hours of PO immunosuppressants (after conversion from IV), consider a less intense level of care setting.

61:

Anti-infective medications may cause adverse drug reactions. Symptoms of an adverse drug reaction include urticaria, angioedema, bronchospasm, pruritus, rash, vomiting, diarrhea, anaphylaxis, and thrombocytopenia.

62:

Guidelines recommend preprandial glucose targets of 80-130 mg/dL (4.4-7.2 mmol/L) and a random blood glucose target range of 100-180 mg/dL (5.6-10.0 mmol/L) in hospitalized patients. In stable patients with previous tight glycemic control, more stringent targets may be appropriate. In patients with severe comorbidities, a history of hypoglycemia, terminal illness, late onset diabetes, and in older patients, less stringent targets may be appropriate. The preferred method for achieving and maintaining glucose control in non-critically ill patients is scheduled subcutaneous insulin administration. Guidelines discourage the sole use of sliding scale insulin in the inpatient hospital setting. Rather, an insulin regimen consisting of basal, prandial, and correctional insulin is recommended in hospitalized patients (American Diabetes Association, Diabetes Care 2024, 47: S1-S314).

63:

The risk of hypoglycemia increases in young children (less than six years of age) who often cannot recognize or communicate their needs or manage hypoglycemia. There is a strong correlation between the frequency of blood glucose monitoring and glucose management. Therefore, confirmation of hypoglycemia may require frequent blood glucose monitoring in young children and infants (American Diabetes Association, Diabetes Care 2024, 47: S1-S314; Abraham et al., Pediatr Diabetes 2018, 19 Suppl 27: 178-92).

64:

Instruction: Insulin adjustments should be based on blood glucose values obtained by lab draw or glucose monitor. Intermittent insulin must be administered at least 3 times in 24 hours. If the patient is on a continuous insulin infusion, the rate must be adjusted at least twice in 24 hours. A dose that is held based on a normal or low blood glucose value should be included in the count of doses administered within the 24-hour period.

65:

Instruction: For these criteria, medication adjustment refers to a change in dose, frequency of administration, or change in medication.

66:

Fluid is replaced based on measured and insensible losses. Examples of measurable fluid losses include urine output, blood loss, nasogastric, wound or ostomy drainage, diarrhea, and vomiting. Examples of insensible losses include perspiration, evaporative losses during surgery, and respirations. In the absence of a direct measurement of losses, such as from young children, daily weight changes can be used to determine fluid losses.

67:

Diagnosing *C. difficile* in the pediatric population is challenging due to the high rate of colonization in the very young and the lack of diagnostic tests validated in this age group. Infants less than 1 year of age demonstrate high rates of asymptomatic colonization. With age, rates of colonization decrease and in children older than 2 years colonization rates approach those found in adults. Testing is only warranted in symptomatic patients due to the high probability of false positive results in asymptomatic patients. Guidelines stress the importance of concomitant investigation of alternative sources of diarrhea (McDonald et al., Clin Infect Dis 2018; Schutze and Willoughby, Pediatrics 2013, 131: 196-200; Sammons et al., JAMA Pediatr 2013, 167: 567-73).

68:

Signs indicating recovery from *Clostridioides difficile* include:

- Decrease in the severity and frequency of diarrhea
- Absence of fever
- Normal white blood cell count

Not all patients demonstrate fever or leukocytosis during the course of illness. Diet advancement should be based on improving clinical signs and symptoms.

69:

Treatment of *C. difficile* in the pediatric population begins with discontinuing any current anti-infectives that may have precipitated the infection as soon as possible. In patients demonstrating fulminant, previously severe, disease or a lack of response to discontinuing the offending anti-infectives, the initiation of antimicrobial therapy specifically targeting *C. difficile* is recommended. Oral vancomycin is considered the first line treatment for most children and adolescents. If the patient is unable to tolerate oral vancomycin or has an ileus, vancomycin may be administered by enema (McDonald et al., Clin Infect Dis 2018; American Academy of Pediatrics. Committee on Infectious Diseases., Red book: Report of the Committee on Infectious Diseases. 2012).

70:

Instruction: This criteria point is intended for anti-infectives initiated for the treatment of *C. difficile* infections. The preferred treatment of *Clostridioides difficile* infections is based on the episode of illness and the severity of the symptoms. The Infectious Diseases Society of America (IDSA) recommends a variety of treatment regimens which are dependent on clinical presentation, including disease severity and how many episodes of disease the patient has experienced (Johnson et al., Clin Infect Dis 2021, 73: e1029-e44).

71:

Inadequate oral intake refers to the inability to maintain hydration according to a patient's documented fluid goal or maintenance fluid requirement. The inability to maintain hydration may occur when excessive output (e.g., vomiting, diarrhea, or insensible losses) leads to a net fluid deficit.

72:

Chest tubes are used to evacuate air, fluid, or blood from the pleural space. For malignancies involving the pleura, chest tubes may also be used for the administration of sclerosing agents. Serial chest x-rays may be performed to ensure that there is no re-accumulation of air or fluid. The chest tube may be removed when there is no evidence of air leak, output has slowed to 250-400 mL/day, and the lung is fully expanded (Vuong et al., Respir Med 2018, 137: 152-66).

73:

The discharge screen is a resource tool and not criteria. Referring to the discharge screen at the initiation of discharge planning is recommended.

74:

The home environment and safety assessment will normally include an evaluation of the patient's pre- and post-hospitalization functional level, physical layout of the home (e.g., entrances and exits, stairs, access to the community), identification of unsafe conditions (e.g., scatter rugs, missing handrails, oxygen use and smoking, lack of fire safety devices, inadequate lighting, heating, and cooling), factors that may trigger symptoms (e.g., secondhand smoke, poor food choices, dysfunctional family dynamics), sanitation hazards (e.g., lack of electricity, running water, refrigeration, inadequate toilet facilities, presence of insects or rodents), and the need for adaptive equipment (e.g., walker, handrails for the tub or shower, elevated toilet seat).

75:

Ensuring the patient and/or caregiver understands all aspects of the condition and can assume responsibility for self-care is crucial in preventing readmission. Assessing a patient and/or caregiver's level of understanding after education has been provided can be difficult. The teach-back method is an effective way of providing education at

the appropriate level and can also be used to assess the learner's comprehension. To use the teach-back method, discuss key points in common terms and avoid using medical jargon or unfamiliar terms. After the education has been delivered, ask the learner to repeat what was learned. Gaps or misinterpretations in the learner's explanation will pinpoint areas where communication may have failed and provide opportunity for clarification.

76:

In order to avoid subsequent pulmonary exacerbations, prevention techniques include vaccination, avoidance of crowds, and frequent hand washing.

77:

Medication reconciliation is a formal process or technique used by health care providers and pharmacists to identify the most complete and accurate list of all medications a patient is taking at times of transitions in care (e.g., upon hospital admission, transfer from one unit to another during hospitalization, or discharge from the hospital to home or another facility). The goals of this process are to ensure medication and dosages are appropriate for the patient, resolve discrepancies in drug regimens, and ultimately prevent medication errors and reduce adverse drug events. Medication reconciliation is a Joint Commission National Patient Safety goal. Coordinating information when a patient is transferred to another setting, service, practitioner, or level of care ensures accurate medications are listed. The process is comprised of the following (Hospital: National Patient Safety Goals. 2020):

- Obtain an external list of medications (e.g., medications taken prior to admission)
- Develop a list of current medications and add them to the medical record
- Compare the medications from the external list to the current list
- Clarify inconsistencies/discrepancies (e.g., omissions, duplications, contraindications, unclear information, and changes)
- Develop a list of medications to be prescribed at discharge or transfer
- Communicate the new list to the patient and/or appropriate caregiver(s)
- Ensure that patient and/or caregiver(s) understand the medication information upon discharge

Specific medication issues identified as being problematic include missing medication information from transfer orders, lack of information on medications provided in the acute setting, incomplete medication records, discrepancies between hospital regimen and discharge summary, and missing information pertaining to the patient's tolerance of a medication regimen. Although medication reconciliation is an important aspect in patient safety, there is a lack of consensus and evidence about the best effective methods of implementing this process. Physician-led and electronic medication reconciliation in hospitals are effective strategies to reduce medication discrepancies, however, the impact of these interventions is uncertain due to the low quality of evidence (Choi and Kim, J Clin Pharm Ther 2019, 44: 932-45; Redmond et al., Cochrane Database Syst Rev 2018, 8: CD010791). Trained pharmacy technicians, under the direction of a licensed pharmacist, may be an option for developing and expanding medication reconciliation processes (Irwin et al., Hosp Pharm 2017, 52: 44-53).

78:

Close follow-up of older adults, pediatric, or at-risk patients after discharge can help minimize hospital readmission and total health care costs (Rising et al., Ann Emerg Med 2013, 62: 145-50; Frei-Jones et al., Pediatr Blood Cancer 2009; 52(4): 481-485; Kripalani et al., JAMA 2007; 297(8): 831-841). Studies have shown that for greater than 50% (0.50) of patients re-hospitalized within 30 days of discharge, there was no visit to a physician's office between the time of discharge and re-hospitalization (Hernandez et al., JAMA 2010, 303: 1716-22; Jencks et al., N Engl J Med 2009; 360(14): 1418-1428). Efforts to reduce readmissions should focus on patients known to be at high risk and patients who have frequent visits to the emergency department (ED). Patients at high risk may have multiple ED visits prior to readmission (Rising et al., Ann Emerg Med 2013, 62: 145-50).

79:

Outpatient management contraindicated refers to the following:

- The patient whose medical condition is so fragile that going to the clinic or medical practitioner's office for treatment would put the patient at risk
- The patient with extensive equipment (e.g., ventilators, specialized beds) where travel outside the home would be cumbersome and places the patient at risk
- The treatment goal is to adapt the home environment for increased independence (e.g., an individual who is blind who needs mobility education)

80:

Outpatient management unavailable refers to the following:

- The patient whose treatment (e.g., dressing changes) is so extensive that it takes too long or is too frequent (e.g., twice daily) to be done in an outpatient clinic or medical practitioner's office
- The scheduling of the treatment falls outside the medical practitioner's office or clinic hours
- Travel distance to receive care is excessive (e.g., greater than 1 hour)
- The service is unavailable in the outpatient setting

81:

Skilled nursing services refer to services that must be provided by a licensed nurse qualified to assess and monitor patient condition(s) and provide medical treatment and/or teaching to patients who have skilled nursing needs. Examples of Home Care level of care interventions include skilled assessment, intervention, or education for any of the following:

- New treatment or medication regimen
- Adjustment in the treatment or medication regimen
- Infusion therapy administration, teaching, or access device management
- Management and evaluation of a care plan for patients with an active comorbidity and multiple care needs
- Chronic disease requiring a disease management program (e.g., asthma, heart failure, chronic obstructive pulmonary disease, diabetes)
- End-stage disease, hospice, or palliative care

82:

Skilled nursing interventions refer to services that are provided or supervised by a qualified or licensed professional when the care needs of the patient require assessment, observation, monitoring, and/or education to achieve the medically desired result. Close observation and assessment by nursing may be necessary at this level of care. This includes nursing care such as medication administration (excludes routine), vital signs monitoring, and/or other interventions not listed.

83:

The Home Care criteria are used for a patient who has been admitted or is expected to be admitted for home care. When criteria are met for this level of care and it is not available in your area, we recommend you use the next higher level of care.

Home care can include specialty services provided by a psychiatric nurse following a specialized treatment plan developed by a psychiatrist or advanced practice registered nurse. It is intended for treatment and monitoring of a patient's psychiatric condition that prevents the patient from leaving the house or makes leaving the house extremely difficult or unsafe (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022). Home care can also be used for a patient with severe and persistent mental illness with a history of nonadherence to medication.

Evaluation and Treatment

Programming may differ based upon legislative and geographical variances and is subject to organizational policy; however, at a minimum, it should include:

- Care coordination with treating physician and other providers
- Clinical assessment every visit
- Education every visit
- Medication assessment, teaching, and management every visit
- Medication reconciliation initiated within first visit
- Psychosocial assessment within first visit
- Substance evaluation within first 2 visits

For those patients who are covered under a Medicare benefit, the Affordable Care Act mandates that a certifying physician or allowed practitioner have a face-to-face visit with the patient within 90 days prior to the start of care, or within 30 days after the start of care. There must be documentation of the patient's condition that supports the need for home care services and the requirement of homebound status (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022).

84:

Support system includes friends, family, social services, health care delivery providers or agencies, sponsors, clergy, and self-help groups.

85:

The initial clinical assessment is performed by a licensed professional, which includes a comprehensive review of the patient's presenting diagnosis and a review of body systems. Identification of current and potential medical needs and health problems are identified. The clinical assessment includes:

- Patient history, physical exam, vital signs, height and weight
- Pain assessment includes: cause, intensity, quality, onset, duration, effects on quality of life with activities of daily living (ADLs), instrumental activities of daily living (IADLs), play, and interpersonal relationships, including a current pain management regimen.
- Patient's nutritional and hydration status
- Effects of current illness or injury on age appropriate function
- Safety and infection control measures
- Prescription and over-the-counter (OTC) medications, herbal supplements, and home remedies
- Patient's and/or caregiver's understanding of medication use, dosing, side effects and adherence to the medication or treatment regimen
- Patient's and/or caregiver's understanding of the illness or disease process and potential long-term complications
- Patient and family's coping strategies to deal with the illness or injury and ability to follow the plan of care

86:

Goal-directed therapy includes one or more of the following:

- Activities of daily living (ADLs) (e.g., feeding, dressing, bathing, grooming, toileting, functional mobility)
- Bed mobility or transfers (e.g., transfers to chair, toilet, commode, tub, shower, car)
- Cognitive, speech, language, or swallowing retraining
- Home and lifestyle modification (e.g., assessment and training focused on the home, job, school, or work environment)
- Pain management techniques (e.g., assessment and interventions used to relieve pain or decrease its intensity)
- Pulmonary rehabilitation (e.g., energy conservation, speech, breathing re-education, postural drainage, percussion, or vibration)
- Positioning and splinting
- Range of motion and strengthening
- Wheelchair mobility
- Ambulation

87:

Rehabilitation potential refers to the probability that therapy and medical goals are realistic and attainable based on patient's prior level of function, severity of illness or injury, and the extent of impairments.

88:

Functional assistance levels are based upon the patient's function during tasks and activities necessary to return to household mobility or ambulation. The functional assistance level required for each individual task or activity (e.g., mobility, activities of daily living (ADLs)) may vary.

The following terms are commonly used in the post-acute setting:

- Independent - Patient can safely and within a reasonable amount of time perform a task (or developmentally appropriate task) without physical or cognitive assistance or supervision
- Modified Independent - Patient performs an activity with a supportive device, adaptive equipment, and/or prosthetic or orthotic device. Additional time may be required to complete the activity and/or there are safety (risk) considerations
- Supervision - Patient performs an activity with standby or distant supervision or setup. Verbal cueing or coaxing, without physical contact or setup of items and application of orthoses may be required when patient's safety awareness is impaired

- Minimum or limited Assistance - Patient performs at least 75%(0.75) of an activity and requires some physical contact to steady, guide, or move
- Moderate or extensive Assistance - Patient performs at least 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Maximum Assistance - Patient performs 25%(0.25) to 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Total Assistance or dependence - Patient performs less than 25%(0.25) of an activity and may require total assistance for functional mobility or ADLs

89:

The intensity of rehabilitation therapies is a driver for admission to an acute inpatient rehabilitation facility. Documentation should reflect the need for required and ongoing intensive therapies. The determination of whether a patient requires and can tolerate an intensive rehabilitation program of at least 15 hours is not based on any one single piece of documentation. All clinical documentation should be evaluated by the reviewer to determine whether the patient is demonstrating the need for services and has the ability to tolerate them. Patients who do not require this level of intensive therapy should be treated at a lower level of care. Rehabilitation therapies should be reasonable and medically appropriate and should not exceed a patient's level of tolerance. If there are clinical features which preclude the patient's ability to tolerate the intensity threshold of at least 15 hours per week, secondary review would be appropriate (Centers for Medicare & Medicaid Services, Medicare Benefit Policy Manual Chapter 1 - Inpatient Hospital Services Covered Under Part A. 2021; Miller and Forer, Pm r 2021, 13: 535-9; Forrest et al., Medicine (Baltimore) 2019, 98: e17096).

90:

Comorbid conditions include new onset or worsening behavioral symptoms, heart failure, deep vein thrombosis, uncontrolled diabetes, immunocompromised states, malignant or end-stage disease, malnutrition, renal insufficiency or end-stage renal disease, infection, ventilator dependence, noninvasive positive pressure ventilation or respiratory insufficiency, complex wound care, and new functional impairment(s) with limitation.

91:

Ventilation includes invasive ventilation and noninvasive positive pressure ventilation (NIPPV), including continuous positive airway pressure (CPAP) and bilevel positive airway pressure (BiPAP).

92:

Notable causes for prolonged mechanical ventilation include:

- Ventilator-acquired pneumonia
- Renal failure
- Heart failure
- Exacerbation of chronic obstructive pulmonary disease (COPD)
- Chest wall disorder
- Neuromuscular diseases
- Traumatic brain injury
- Sepsis
- Complications following cardiac surgical procedures

Major contributing factors for failure to wean include (Ambrosino and Vitacca, Multidiscip Respir Med 2018, 13: 6; Davies et al., Br J Anaesth 2017, 118: 563-9):

- Increased work of breathing
- Impaired respiratory drive
- Inspiratory muscle weakness

93:

Long-Term Acute Care (LTAC) is a recognized level of care designation by the Centers for Medicare and Medicaid Services (CMS) for acute care hospitals. It is defined by federal statutes as a level of care for patients requiring an average length of stay of 25 days or more (Centers for and Medicaid Services, Fed Regist 2018, 83: 41144-784). Patients who require a short stay for treatment may be more appropriate for the subacute or skilled nursing facility (SNF) level of care.

